

Question	Answer	Marks
1(a)(i)	force per unit mass	B1
1(a)(ii)	acceleration = F/m , field strength = F/m , so equal	B1
1(b)	smooth curve between R and $4R$ with negative gradient of decreasing magnitude	B1
	line passing through $(R, 1.00g)$ and $(2R, 0.25g)$	B1
	line ending at $(4R, 0.0625g)$	B1
1(c)	$M = (4/3 \times \pi R^3)\rho$	C1
	$g = GM/(2R)^2$	C1
	$g = \frac{1}{3} \times 6.67 \times 10^{-11} \times \pi \times 3.4 \times 10^6 \times 4.0 \times 10^3$ $= 0.95 \text{ ms}^{-2}$	A1

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Question	Answer	Marks
2(c)	graph that shows a variation of V with constant T	M1